33.4: Ecdysozoans are the most species-rich animal group

An evolutionary tree highlights ecdysozoa.

Branch point 1: First level-1 branch: Porifera. The second level-1 branch splits from branch point 2 into the first level-2 branch: Cnidaria. The second level-2 branch splits from branch point 3 into the first and second level-3 branches. The first level-3 branch splits from branch point 4 into the first level-4 branch: Lophotrochozoa and the second level-4 branch: Ecdysozoa. The second level-3 branch is Deuterostomia.

Although defined primarily by molecular evidence, the clade Ecdysozoa includes animals that shed a **cuticle** (a tough external coat) as they grow. In fact, the group derives its name from this process, which is called *ecdysis*, or **molting**. Ecdysozoa includes about eight animal phyla and contains more known species than all other animal, protist, fungus, and plant groups combined. Here we'll focus on the two largest ecdysozoan phyla, the nematodes and arthropods, which are among the most successful and abundant of all animal groups.

Nematodes

Among the most ubiquitous of animals, nematodes (phylum Nematoda), or roundworms, are found in most aquatic habitats, in the soil, in the moist tissues of plants, and in the body fluids and tissues of animals. The cylindrical bodies of nematodes range from less than 1mm to more than 1m long, often tapering to a fine tip at the posterior end and to a blunter tip at the anterior end **(Figure 33.26)**. A nematode's body is covered by a tough cuticle (a type of exoskeleton); as the worm grows, it periodically sheds its old cuticle and secretes a new, larger one. Nematodes have an alimentary canal, though they lack a circulatory system. Nutrients are transported throughout the body via fluid in the hemocoel. The body wall muscles are all longitudinal, and their contraction produces a thrashing motion.

Figure 33.26

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A free-living nematode.
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Nematode with a long, thin, whiplike body.

(colorized SEM)

Multitudes of nematodes live in moist soil and in decomposing organic matter on the bottoms of lakes and oceans. While 25,000 species are known, perhaps 20 times that number actually exist. These free-living worms play an important role in decomposition and nutrient cycling, but little is known about most species. One species of soil nematode, *Caenorhabditis elegans*, however, is very well studied and has become a model research organism in biology (see Concept 47.3). Ongoing studies of *C. elegans* are providing insight into mechanisms involved in aging in humans, as well as many other topics.

Phylum Nematoda includes many species that parasitize plants, and some are major agricultural pests that attack the roots of crops. Other nematodes parasitize animals. Some of these species benefit humans by attacking insects such as cutworms that feed on the roots of crop plants. On the other hand, humans are hosts to at least 50 nematode species, including various pinworms and hookworms. One notorious nematode is *Trichinella spiralis*, the worm that causes trichinosis **(Figure 33.27)**. Humans acquire this nematode by eating raw or undercooked pork or other meat (including wild game such as bear or walrus) that has juvenile worms encysted in the muscle tissue. Within the human intestines, the juveniles develop into sexually mature adults. Females burrow into the intestinal muscles and produce more juveniles, which bore through the body or travel in lymphatic vessels to other organs, including skeletal muscles, where they encyst.

Figure 33.27

```
    Juveniles of the parasitic nematode *Trichinella spiralis*
encysted in human muscle tissue.
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Cysts (round structures) within human muscle tissue contain multiple worm-shaped parasitic nematodes. Scale: 50 micrometers.

(LM)

Parasitic nematodes have an extraordinary molecular toolkit that enables them to redirect some of the cellular functions of their hosts. Some species inject their plant hosts with molecules that induce the development of root cells, which then supply nutrients to the parasites. When *Trichinella* parasitizes animals, it regulates the expression of specific muscle cell genes encoding proteins that make the cell elastic enough to house the nematode. Additionally, the infected muscle cell releases signals that promote the growth of new blood vessels, which then supply the nematode with nutrients.

Arthropods

Zoologists estimate that there are about a billion billion (1018) arthropods living on Earth. More than 1 million arthropod species have been described, most of which are insects. In fact, two out of every three

known species are arthropods, and members of the phylum Arthropoda can be found in nearly all habitats of the biosphere. By the criteria of species diversity, distribution, and sheer numbers, arthropods must be regarded as the most successful of all animal phyla.

Arthropod Origins

Biologists hypothesize that the diversity and success of **arthropods** are related to their body plan—their segmented body, hard exoskeleton, and jointed appendages. How did this body plan arise and what advantages did it provide?

The earliest fossils of arthropods are from the Cambrian explosion (535–525 millionyears ago), indicating that the arthropods are at least that old. The fossil record of the Cambrian explosion also contains many species of *lobopods*, a group from which arthropods may have evolved. Lobopods such as *Hallucigenia* (see **Figure 32.7**) had segmented bodies, but most of their body segments were identical to one another. Early arthropods, such as the trilobites, also showed little variation from segment to segment **(Figure 33.28)**. As arthropods continued to evolve, groups of segments tended to become functionally united into "body regions" specialized for tasks such as feeding, walking, or swimming. These evolutionary changes resulted not only in great diversification but also in efficient body plans that permit the division of labor among different body regions.

Figure 33.28

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A trilobite fossil.
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A trilobite fossil has a body that is roughly tear-shaped with many obvious segments and spiny structures lining one end.

Trilobites were common denizens of the shallow seas throughout the Paleozoic era but disappeared with the great Permian extinctions about 250 millionyears ago. Paleontologists have described about 4,000 trilobite species.

What genetic changes led to the increasing complexity of the arthropod body plan? Arthropods today have two unusual *Hox* genes, both of which influence segmentation. To test whether these genes could have driven the evolution of increased body segment diversity in arthropods, researchers studied *Hox* genes in onychophorans (see Figure 33.2), close relatives of arthropods **(Figure 33.29)**. Their results indicate that the diversity of arthropod body plans did *not* arise from the acquisition of new *Hox* genes. Instead, the evolution of body segment diversity in arthropods was probably driven by changes in the sequence or regulation of existing *Hox* genes (see Concept 25.5).

Figure 33.29

Inquiry: Did the arthropod body plan result from new *Hox*genes?

Experiment

One hypothesis suggests that the arthropod body plan resulted from the origin (by gene duplication and subsequent mutations) of two unusual *Hox* genes found in arthropods: *Ultrabithorax* (*Ubx*) and *abdominal-A* (*abd-A*). Researchers tested this hypothesis using onychophorans, a group of invertebrates closely related to arthropods. Unlike many living arthropods, onychophorans have a body plan in which most body segments are identical to one another. If the origin of the *Ubx* and *abd-A Hox* genes drove the evolution of body segment diversity in arthropods, these genes probably arose on the arthropod branch of the evolutionary tree:

A phylogenetic tree. A branch begins with a common ancestor of onychophorans and arthropods. The branch that goes to arthropods has a point labeled Origin of U b x and a b d, A Hox genes?.

According to this hypothesis, *Ubx* and *abd-A* would not have been present in the common ancestor of arthropods and onychophorans; hence, onychophorans should not have these genes. The researchers examined the *Hox* genes of the onychophoran *Acanthokara kaputensis*.

Results

The onychophoran *A. kaputensis* has all arthropod *Hox* genes, including *Ubx* and *abd-A*.

A micrograph of a onychophoran embryo. Antenna, jaws, and L 1 through 15 body segments are highlighted. Red in L 15 indicates the body regions of this onychophoran embryo in which U b x or a b d, A genes were expressed.

Conclusion

The evolution of increased body segment diversity in arthropods was not related to the origin of new *Hox* genes.

Data from J. K. Grenier et al., Evolution of the entire arthropod *Hox* gene set predated the origin and radiation of the onychophoran/arthropod clade, *Current Biology* 7:547–553 (1997).

WHAT IF? *Suppose*A. kaputensis*did not have the*Ubx*and*abd-A*Hox*genes. How would the conclusions of this study have been affected? Explain.*

*For suggested answer, see*Appendix A*.*

General Characteristics of Arthropods

Over the course of evolution, the appendages of some arthropods have become modified, specializing in functions such as walking, feeding, sensory reception, reproduction, and defense. Like the appendages from which they were derived, these modified structures are jointed and come in pairs. **Figure 33.30** illustrates the diverse appendages and other arthropod characteristics of a lobster.

Figure 33.30

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External anatomy of an arthropod.
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A diagram of a lobster.

Many of the distinctive features of arthropods are apparent in this dorsal view of a lobster. The body is segmented, but this character is obvious only in the post-genital region or "tail," located behind the genitals. The appendages (including antennae, pincers, mouthparts, walking legs, and swimming appendages) are jointed. The head bears a pair of compound (multilens) eyes. The whole body, including appendages, is covered by an exoskeleton.

Major body segments include the cephalothorax (head with eyes and thorax) and the abdomen. Four walking legs are visible from the side view, along with swimming appendages (one pair per body segment), pincers (for defense), and antennae (for sensory reception).

The body of an arthropod is completely covered by the cuticle, an exoskeleton constructed from layers of protein and the polysaccharide chitin. As you know if you've ever eaten a crab or lobster, the cuticle can be thick and hard over some parts of the body and thin and flexible over others, such as the joints. The rigid exoskeleton protects the animal and provides points of attachment for the muscles that move the appendages. But it also prevents the arthropod from growing, unless it occasionally sheds its exoskeleton and produces a larger one. This molting process is energetically expensive, and it leaves the arthropod vulnerable to predation and other dangers until its new, soft exoskeleton hardens.

When the arthropod exoskeleton first evolved in the sea, its main functions were probably protection and anchorage for muscles, but it later enabled certain arthropods to live on land. The exoskeleton's relative impermeability to water helped prevent desiccation, and its strength provided support when arthropods left the buoyancy of water. Fossil evidence suggests that arthropods were among the first animals to colonize

land, roughly 450 millionyears ago. These fossils include fragments of arthropod remains, as well as possible millipede burrows. Arthropod fossils from several continents indicate that by 410 millionyears ago, millipedes, centipedes, spiders, and a variety of wingless insects all had colonized land.

Arthropods have well-developed sensory organs, including eyes, olfactory (smell) receptors, and antennae that function in both touch and smell. Most sensory organs are concentrated at the anterior end of the animal, although there are interesting exceptions. Female butterflies, for example, “taste” plants using sensory organs on their feet.

Like many molluscs, arthropods have an **open circulatory system**, in which fluid called *hemolymph* is propelled by a heart through short arteries and then into the hemocoel—the body cavity surrounding the tissues and organs. (The term *blood* is generally reserved for fluid in a closed circulatory system.) Hemolymph reenters the arthropod heart through pores that are usually equipped with valves. In most arthropods, the coelom that forms in the embryo becomes much reduced as development progresses, and the hemocoel becomes the main body cavity in adults.

A variety of specialized gas exchange organs have evolved in arthropods. These organs allow the diffusion of respiratory gases in spite of the exoskeleton. Most aquatic species have gills with thin, feathery extensions that expose a large surface area to the surrounding water. Terrestrial arthropods generally have internal surfaces specialized for gas exchange. For example, most insects have tracheal systems, branched air ducts leading into the interior of the body from pores in the cuticle.

Morphological and molecular data suggest that living arthropods consist of three major lineages that diverged early in the phylum’s evolution: **chelicerates** (sea spiders, horseshoe crabs, scorpions, ticks, mites, and spiders); **myriapods** (centipedes and millipedes); and **pancrustaceans** (a recently defined, diverse group that includes insects as well as lobsters, shrimp, barnacles, and other crustaceans).

Chelicerates

Chelicerates (clade Chelicerata) are named for clawlike feeding appendages called **chelicerae**, which serve as pincers or fangs. Chelicerates lack antennae, and most have simple eyes (eyes with a single lens).

The earliest chelicerates were **eurypterids**, or water scorpions. These marine and freshwater predators grew up to 3m long; it is thought that some species could have walked on land, much as land crabs do today. Most of the marine chelicerates, including all of the eurypterids, are extinct. Among the marine chelicerates that survive today are the sea spiders (pycnogonids) and horseshoe crabs **(Figure 33.31)**.

Figure 33.31

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Horseshoe crabs (*Limulus polyphemus*).
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Horseshoe crabs on a beach.

Common on the Atlantic and Gulf coasts of the United States, these "living fossils" have changed little in hundreds of millions of years. They are surviving members of a rich diversity of chelicerates that once filled the seas.

The bulk of modern chelicerates are **arachnids**, a group that includes scorpions, spiders, ticks, and mites **(Figure 33.32)**. Nearly all ticks are bloodsucking parasites that live on the body surfaces of reptiles or mammals. Parasitic mites live on or in a wide variety of vertebrates, invertebrates, and plants.

Figure 33.32

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Arachnids.
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Scorpion has coiled tail and extended pincers. Dust mite has dome-like body (scale: 50 micrometers). Spider with bright coloration is on a web.

Scorpions have pedipalps that are pincers specialized for defense and the capture of food. The tip of the tail bears a poisonous stinger. Dust mites are ubiquitous scavengers in human dwellings but are harmless except to those people who are allergic to them (colorized SEM). Web-building spiders are generally most active during the daytime.

Arachnids have six pairs of appendages: the chelicerae; a pair of appendages called *pedipalps* that function in sensing, feeding, defense, or reproduction; and four pairs of walking legs. Spiders use their fang-like chelicerae, which are equipped with poison glands, to attack prey. As the chelicerae pierce the prey, the spider secretes digestive juices onto the prey's torn tissues. The food softens, and the spider sucks up the liquid meal. In most spiders, gas exchange is carried out by **book lungs**, stacked platelike structures contained in an internal chamber **(Figure 33.33)**. The extensive surface area of these respiratory organs enhances the exchange of O2 and CO2 between the hemolymph and air.

Figure 33.33

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Book lungs.
```

Spider body plan. Pedipalps, chelicera, heart and book lungs are highlighted.

A unique adaptation of many spiders is the ability to catch insects by constructing webs of silk, a liquid protein produced by specialized abdominal glands. The silk is spun by organs called spinnerets into fibers that then solidify. Each spider engineers a web characteristic of its species and builds it perfectly on the first try, indicating that this complex behavior is inherited. Various spiders also use silk in other ways: as droplines for rapid escape, as a cover for eggs, and even as “gift wrap” for food that males offer females during courtship. Many small spiders also extrude silk into the air and let themselves be transported by wind, a behavior known as “ballooning.”

Myriapods

Millipedes and centipedes belong to the clade Myriapoda **(Figure 33.34)**. All living myriapods are terrestrial. The myriapod head has a pair of antennae and three pairs of appendages modified as mouthparts, including the jaw-like mandibles.

Figure 33.34

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Myriapods.
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Part a. Millipede: thin black segmented body with legs extending from each segment. Part b. Centipede: brownish segmented body, shorter and with fewer legs than the millipede.

Millipedes have a large number of legs, though fewer than the thousand their name implies. Each trunk segment is formed from two fused segments and bears two pairs of legs (see Figure 33.34a). Millipedes eat decaying leaves and other plant matter. They may have been among the earliest animals on land, living on mosses and early vascular plants.

Unlike millipedes, centipedes are carnivores. Each segment of a centipede’s trunk region has one pair of legs (see Figure 33.34b). Centipedes have poison claws on their foremost trunk segment that paralyze prey and aid in defense.

Pancrustaceans

A series of recent papers present evidence that terrestrial insects are more closely related to lobsters and other crustaceans than they are to the terrestrial group we just discussed, the myriapods (millipedes and centipedes). These studies also suggest that the diverse group of organisms referred to as crustaceans are paraphyletic: Some lineages of crustaceans are more closely related to insects than they are to other crustaceans **(Figure 33.35)**. However, together the insects and crustaceans form a clade, which

systematists have named Pancrustacea (from the Greek *pan*, all). We turn next to a description of the members of Pancrustacea, focusing first on crustaceans and then on the insects.

Figure 33.35

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The phylogenetic position of the insects.
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Tree diagram for the phylogenetic position of the insects.

Recent results have shown that the insects are nested within lineages of aquatic crustaceans. The remipedians are one of several groups of aquatic crustaceans that may be the sister group to the insects.

Tree diagram represented as a list.

Common ancestor of arthropods (branch point 1).

Branch point 2.

Branch point 3.

Branch point 4. Insects (example: beetle). (dotted line to) Remipedians (a crustacean group). Other crustaceans (example: lobster). Myriapods (example: centipede). Chelicerates (example: scorpion)

DRAW IT *Circle the portions of this tree that comprise the clade Pancrustacea.*

*For suggested answer, see*Appendix A.**

Crustaceans

Crustaceans (crabs, lobsters, shrimps, barnacles, and many others) thrive in a broad range of marine, freshwater, and terrestrial environments. Many crustaceans have highly specialized appendages. Lobsters and crayfishes, for instance, have a toolkit of 19 pairs of appendages (see **Figure 33.30**). The anterior-most appendages form two pairs of antennae; crustaceans are the only arthropods with two pairs. Three or more pairs of appendages are modified as mouthparts, including the hard mandibles. Walking legs are present on the thorax, and, unlike their terrestrial relatives, the insects, crustaceans also have appendages on their post-genital region, or “tail.”

Small crustaceans exchange gases across thin areas of the cuticle; larger species have gills. Nitrogenous wastes also diffuse through thin areas of the cuticle, but a pair of glands regulates the salt balance of the hemolymph.

Sexes are separate in most crustaceans. In the case of lobsters and crayfishes, the male uses a specialized pair of abdominal appendages to transfer sperm to the reproductive pore of the female during copulation.

Most aquatic crustaceans go through one or more swimming larval stages.

One of the largest groups of crustaceans (numbering over 11,000 species) is the *isopods*, which include terrestrial, freshwater, and marine species. Some isopod species are abundant in habitats at the bottom of the deep ocean. Among the terrestrial isopods are the pill bugs, or wood lice, common on the undersides of moist logs and leaves.

Lobsters, crayfishes, crabs, and shrimps are all relatively large crustaceans called *decapods***(Figure 33.36)**. The cuticle of decapods is hardened by calcium carbonate. Most decapod species are marine. Crayfishes, however, live in fresh water, and some tropical crabs live on land.

Figure 33.36

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A ghost crab, an example of a decapod.
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Ghost crab is a pale brownish-white color.

Ghost crabs live on sandy ocean beaches worldwide. Primarily nocturnal, they take shelter in burrows during the day.

Many small crustaceans are important members of marine and freshwater plankton communities. Planktonic crustaceans include many species of *copepods*, which are among the most numerous of all animals. Some copepods are grazers that feed upon algae, while others are predators that eat small animals (including smaller copepods!). Copepods are rivaled in abundance by the shrimplike krill, which grow to about 5cm long **(Figure 33.37)**. A major food source for baleen whales (including blue whales, humpbacks, and right whales), krill are now being harvested in great numbers by humans for food and agricultural fertilizer. The larvae of many larger-bodied crustaceans are also planktonic.

Figure 33.37

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Krill.
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A krill swims.

These planktonic crustaceans are consumed in vast quantities by some whales.

With the exception of a few parasitic species, barnacles are a group of sessile crustaceans whose cuticle is hardened into a shell containing calcium carbonate **(Figure 33.38)**. Most barnacles anchor themselves to rocks, boat hulls, pilings, and other submerged surfaces. Their natural adhesive is as strong as synthetic glues. These barnacles feed by extending appendages from their shell to strain food from the water.

Barnacles were not recognized as crustaceans until the 1800s, when naturalists discovered that barnacle larvae resemble the larvae of other crustaceans. The remarkable mix of unique traits and crustacean homologies found in barnacles was a major inspiration to Charles Darwin as he developed his theory of evolution.

Figure 33.38

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Barnacles.
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Small appendages extend from a barnacle.

The jointed appendages projecting from the barnacles' shells capture organisms and organic particles suspended in the water.

We turn now to a group nested within the paraphyletic crustaceans, the insects.

Insects

Insects and their six-legged terrestrial relatives form an enormous clade, Hexapoda; we'll focus here on the insects, since as a group they have more described species than all other eukaryotic groups combined. Insects live in almost every terrestrial habitat and in fresh water, and flying insects fill the air. Insects are rare, though not absent, in marine habitats. The internal anatomy of an insect includes several complex organ systems, which are highlighted in **Figure 33.39**.

Figure 33.39

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Anatomy of a grasshopper, an insect.
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Grasshopper diagram illustrates insect body plan.

The insect body has three regions: head, thorax, and post-genital region. The segmentation of the thorax and post-genital region is obvious, but the segments that form the head are fused.

Parts are labeled as follows.

Cerebral ganglia. The two nerve cords meet in the head, where the ganglia of several anterior segments are fused into a brain (colored white below). The antennae, eyes, and other sense organs are concentrated on the head. Insect mouthparts are formed from several pairs of modified appendages. The mouthparts include mandibles, which grasshoppers use for chewing. In other insects, mouthparts are specialized for lapping, piercing, or sucking.

Nerve cords. The insect nervous system consists of a pair of ventral nerve cords with several segmental ganglia.

Tracheal tubes. Gas exchange in insects is accomplished by a tracheal system of branched, chitin-lined tubes that infiltrate the body and carry oxygen directly to cells. The tracheal system opens to the outside of the body through spiracles, pores that can control air flow and water loss by opening or closing.

Heart. The insect heart drives hemolymph through an open circulatory system.

Malpighian tubules. Metabolic wastes are removed from the hemolymph by excretory organs called Malpighian tubules, which are outpocketings of the digestive tract.

A grasshopper diagram illustrates insect body plan. Head with compound eyes and antennae is anterior to thorax, which is anterior to abdomen. Internally, beginning in the head, the cerebral ganglion, insect mouthparts, crop, ovary, vagina and anus are also labeled. Nerve cords run the length of the body, as do vessels (dorsal artery is labeled, which connects to the heart). Malpighian tubules branch in the posterior region. Tracheal tubes labeled near center of body.

The oldest insect fossils date to about 415 millionyears ago. Later, an explosion in insect diversity took place when insect flight evolved during the Carboniferous and Permian periods (359–252 millionyears ago). An animal that can fly can escape predators, find food and mates, and disperse to new habitats more effectively than an animal that must crawl about on the ground. Many insects have one or two pairs of wings that emerge from the dorsal side of the thorax. Because the wings are extensions of the cuticle, insects can fly without sacrificing any walking legs **(Figure 33.40)**. By contrast, the flying vertebrates—birds and bats—have one of their two pairs of walking legs modified into wings, making some of these species clumsy on the ground.

Figure 33.40

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Ladybird beetle in flight.
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Photograph shows both wings and some legs extended while ladybird beetle in flight.

Insects also radiated in response to the origin of new plant species, which provided new sources of food. By the speciation mechanisms described in Concept 24.2, an insect population feeding on a new plant species can diverge from other populations, eventually forming a new species of insect. A fossil record of diverse insect mouthparts, for example, suggests that specialized modes of feeding on gymnosperms and other Carboniferous plants contributed to early adaptive radiations of insects. Later, a major increase in insect diversity appears to have been stimulated by the evolutionary expansion of flowering plants during the mid-Cretaceous period (about 100 millionyears ago). Although insect and plant diversity decreased during the

Cretaceous mass extinction, both groups have rebounded over the past 66 millionyears. Increases in the diversity of particular insect groups have often been associated with radiations of the flowering plants on which they fed.

Many insects undergo metamorphosis during their development. In the **incomplete metamorphosis** of grasshoppers and some other insect groups, the young (called nymphs) resemble adults but are smaller, have different body proportions, and lack wings. The nymph undergoes a series of molts, each time looking more like an adult. With the final molt, the insect reaches full size, acquires wings, and becomes sexually mature. Insects with **complete metamorphosis** have larval stages specialized for eating and growing that are known by such names as caterpillar, maggot, or grub. The larval stage looks entirely different from the adult stage, which is specialized for dispersal and reproduction. Metamorphosis from the larval stage to the adult occurs during a pupal stage **(Figure 33.41)**.

Figure 33.41

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Complete metamorphosis of a butterfly.
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The stages of becoming a butterfly are a larva (caterpillar), b pupa, c later-stage pupa, (d) emerging adult, and (e) adult.

(a) *The larva (caterpillar) spends its time eating and growing, molting as it grows.* ***(b)*** *After several molts, the larva develops into a pupa.* ***(c)*** *Within the pupa, the larval tissues are broken down, and the adult is built by the division and differentiation of cells that were quiescent in the larva.* ***(d)*** *Eventually, the adult begins to emerge from the pupal cuticle.* ***(e)*** *Hemolymph is pumped into veins of the wings and then withdrawn, leaving the hardened veins as struts supporting the wings. The insect will fly off and reproduce, deriving much of its nourishment from the food reserves stored by the feeding larva.*

Video

Butterfly Emerging

image

Play

image

Play

Reproduction in insects is usually sexual, with separate male and female individuals. Adults come together and recognize each other as members of the same species by advertising with bright colors (as in butterflies),

sounds (as in crickets), or odors (as in moths). Fertilization is generally internal. In most species, sperm are deposited directly into the female's vagina at the time of copulation, though in some species the male deposits a sperm packet outside the female, and the female picks it up. An internal structure in the female called the spermatheca stores the sperm, usually enough to fertilize more than one batch of eggs. Many insects mate only once in a lifetime. After mating, a female often lays her eggs on an appropriate food source where the next generation can begin eating as soon as it hatches.

Insects are classified in more than 30 orders, 8 of which are introduced in **Figure 33.42**.

Figure 33.42

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Exploring Insect Diversity
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Exploring insect diversity.

Although there are more than 30 orders of insects, we'll focus on just 8 here. Two orders of wingless insects, the bristletails (Archaeognatha) and silverfish (Zygentoma), diverged from other insects early in insect evolution. Evolutionary relationships among the other groups discussed here are under debate and so are not depicted on the tree. Tree diagram represented as a list.

Branch point 1. Archaeognatha (bristletails; 350 species). Example is a photograph of a wingless insect on wood.

Branch point 2. Zygentoma (silverfish; 450 species). Example is a wingless insect with long antennae. Winged insects (many orders).

Complete Metamorphosis. Winged insects in this category include the Coleoptera (male snout weevil with very long mouthpart shown); the Diptera (red tachinid shown, a fly-like insect); the Hymenoptera (wasp pictured); and the Lepidoptera (hawkmoth that strongly resembles a hummingbird with long proboscis pictured).

Coleoptera (beetles; 350,000 species): Beetles, such as this male snout weevil (Rhiastus lasternus), constitute the most species-rich order of insects. They have two pairs of wings, one of which is thick and stiff, the other membranous. They have an armored exoskeleton and mouthparts adapted for biting and chewing.

Diptera (151,000 species): Dipterans have one pair of wings; the second pair has become modified into balancing organs called halteres. Their mouthparts are adapted for sucking, piercing, or lapping. Flies and mosquitoes are among the best-known dipterans, which live as scavengers, predators, and parasites. Like many other insects, flies such as this red tachinid (Adejeania vexatrix) have well-developed compound eyes that provide a wide-angle view and excel at detecting fast movements.

Hymenoptera (125,000 species): Most hymenopterans, which include ants, bees, and wasps, are highly social insects. They have two pairs of membranous wings, a mobile head, and chewing or sucking mouthparts. The females of many species have a posterior stinging organ. Many species, such as this European paper wasp (Polistes dominulus), build elaborate nests.

Lepidoptera (120,000 species: Butterflies and moths have two pairs of wings covered with tiny scales. To feed, they uncoil a long proboscis, visible in this photograph of a hummingbird hawkmoth (Macroglossum stellatarum). This moth's name refers to its ability to hover in the air while feeding from a flower. Most lepidopterans feed on nectar, but some species feed on other substances, including animal blood or tears.

Incomplete metamorphosis. Winged insects in this category include the Hemiptera (adult stink bug guarding a cluster of small offspring pictured); and the Orthoptera (spear-bearer katydid pictured; appendages are covered in spiny points and the face has three striking white dots and a very large mouthpart and smaller protrusion from the top of the head, both of which resemble flowers).

Hemiptera (85,000 species): Hemipterans include so-called true bugs, such as stink bugs, bed bugs, and assassin bugs. (Insects in other orders are sometimes erroneously called bugs.)

Hemipterans have two pairs of wings, one pair partly leathery, the other pair membranous. They have piercing or sucking mouthparts and undergo incomplete metamorphosis, as shown in this image of an adult stink bug guarding its offspring (nymphs).

Orthoptera (13,000 species): Grasshoppers, crickets, and their relatives are mostly herbivorous. They have large hind legs adapted for jumping, two pairs of wings (one leathery, one membranous), and biting or chewing mouthparts. This aptly named spiny devil katydid (Panacanthus cuspidatus) has a face and legs specialized for making a threatening display. Male orthopterans commonly make courtship sounds by rubbing together body parts, such as ridges on their hind legs.

Animals as numerous, diverse, and widespread as insects are bound to affect the lives of most other terrestrial organisms, including humans. Insects consume enormous quantities of plant matter; play key roles as predators, parasites, and decomposers; and are an essential source of food for larger animals such as lizards, rodents, and birds. Humans depend on bees, flies, and many other insects to pollinate crops and orchards. In addition, people in many parts of the world eat insects as an important source of protein. On the other hand, insects are carriers for many diseases, including African sleeping sickness (spread by tsetse flies that carry the protist *Trypanosoma*; see **Figure 28.9**) and malaria (spread by mosquitoes that carry the protist *Plasmodium*; see **Figure 23.18** and **Figure 28.18**).

Insects also compete with humans for food. In parts of Africa, for instance, insects claim about 75% of the crops. In the United States, billions of dollars are spent each year on pesticides. But insect populations often evolve pesticide resistance after short periods of time. Try as we may, not even humans have challenged the

preeminence of insects and their arthropod kin. As one prominent entomologist put it: “Bugs are not going to inherit the Earth. They own it now. So we might as well make peace with the landlord.”